

[4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- 3 A light elastic string has natural length a and modulus of elasticity $12mg$. One end of the string is attached to a fixed point O . The other end of the string is attached to a particle of mass m . The particle hangs in equilibrium vertically below O . The particle is pulled vertically down and released from rest with the extension of the string equal to e , where $e > \frac{1}{3}a$. In the subsequent motion the particle has speed $\sqrt{2ga}$ when it has ascended a distance $\frac{1}{3}a$.

Find e in terms of a . [6]

[illegible]